

Semyon Fox

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Builder, swimmer, filmmaker.

2nd-year CS at Galway (First Class Honours). Auditor turned Treasurer of CompSoc, a 450+ member computing society. Builds: React/Next.js frontends, Rust APIs, self-hosted infra.

Technical Skills

Languages	JavaScript, TypeScript, Java, C, SQL, Python, Rust
Web	React, Next.js, Astro, Node.js/Express, REST APIs, HTML/CSS, Tailwind CSS
Databases	PostgreSQL, TimescaleDB, MySQL, Redis
Infrastructure	Docker, Jenkins CI/CD, Git/GitHub, Linux, Nginx, Cloudflare (Workers, Tunnels, Zero Trust), AWS (Fargate, RDS, S3, SES/SQS, IAM), Btrfs, NFS

Education

University of Galway	Galway, Ireland
Bachelor of Science (Honours) in Computer Science and Information Technology	Expected: August 2028

Year 2, with Year 1 First Class Honours.

Modules: Software Engineering, Database Systems (SQL), OOP (Java), Data Structures & Algorithms, Networks & Data Communication, Computer Systems & Organisation, Digital Security & Cryptography, Modelling (Python/NumPy/Matplotlib), Discrete Mathematics, Linear Algebra, Statistics

Honours & Awards

- **Best Intervarsity Award, CompSoc “Capture the Flag”** Mar 2025, Mar 2026
University of Galway Societies Awards 2025 and 2026; BICS National Society Award 2025 (nominated 2026)
- **Brian Ó Maoilchiaráin Award** Jun 2024
Outstanding Student, Leaving Certificate year, Coláiste an Eachréidh
- **STEM Award, GRETB** Jun 2024

Work Experience

University of Galway Computer Society (CompSoc)	Galway, Ireland
Treasurer – Volunteer	Nov 2024–Present

- Led 450+ member student society across finance, communications, and event strategy
- Organised CompSoc CTF 2026 (110+ participants), securing four corporate sponsors (Evernorth, Siren, Centripetal Networks, LibertyIT) and university grant funding
- Reduced participant costs by 50%; increased event profit
- Fixed CI/CD pipeline and JSX syntax bugs on the compsoc.ie React/TypeScript frontend
- Held three committee roles: Public Relations Officer (Nov 2024–Feb 2025), Auditor (Feb 2025–Mar 2026), Treasurer (Mar 2026–Present)

Computer Repair Technician	Athenry & Galway
Cahill Computers (part-time, 8 months) Lapteck (TY placement, 2023)	

- Hardware upgrades, screen and battery replacements, OS installation, drive cloning, and diagnostics

Projects

OghmaNotes

MVP Deployed

Next.js, TypeScript, PostgreSQL (pgvector), Redis, AWS, Docker

- Markdown e-learning platform (3-person team) with RAG search, quizzing, and FSRS spaced repetition for university students
- Canvas LMS integration with PDF extraction and embedding pipeline for automatic course material import
- Migrated from AWS (S3, RDS, ElastiCache, Fargate) to self-hosted on-prem (pgvector, RustFS) to cut costs
- Live at oghmanotes.ie

Uisce

In Development, targeting Aug 2026

React 19, Node.js/Express, PostgreSQL, Redis, Docker, Jest

- Full-stack platform with role-based access for swimmers, coaches, and committee members
- 58-table PostgreSQL schema (5 logical schemas) covering attendance, meet results, training schedules, squad analytics, equipment

Canvas MCP Server

Open Source

TypeScript, Model Context Protocol SDK, Zod

- MCP server exposing the full Canvas LMS REST API to AI assistants across 15 domains (courses, assignments, grades, etc.)
- Merged and normalised 12 open-source Canvas MCP projects; on github.com/semyonfox/canvas-mcp

Irish Rail Data Pipeline

Running 24/7

Python (asyncio/aiohttp), TimescaleDB, Rust (axum), Docker

- Polls Irish Rail API every 3 seconds, storing train positions and station data in TimescaleDB
- Rust (axum) API serving a live map and delay-tracking dashboard

Home Lab & CI/CD Infrastructure

Ongoing

Docker, Jenkins, Nginx, Cloudflare Tunnels, Btrfs, NFS, Pi-hole

- Self-host 30+ services (Jellyfin, Immich, Vaultwarden, Firefly III, n8n, etc.) in 54 containers, behind a router-level VPN
- 6 Jenkins pipelines auto-deploying OghmaNotes, Uisce, Portfolio, etc. on GitHub push
- Cloudflare Zero Trust tunnels (no open ports for public access); Nginx reverse proxy across 21 vhosts internally
- GFS backup strategy (7 daily, 4 weekly, 12 monthly, yearly-forever) to NAS (RAID) via NFS4 with Btrfs snapshots

Portfolio Website

semyon.ie

Astro, Preact, Tailwind CSS v4, Rust (axum), Docker, Jenkins

- More projects, write-ups, and experiments than fit on a CV
- Astro/Preact frontend with Rust (axum) API backend; auto-deployed via Jenkins CI/CD on push
- AI chatbot answering questions about me, with a system prompt built from data scraped across my own social profiles, served from the Rust backend

Interests

Swimming: competitive pool swimmer with regular training and meets.

Video Production: colour grading, VFX, and editing in DaVinci Resolve for short films and personal projects.

Woodworking: hand-built live-edge furniture pieces.